A Reinforcement Learning Architecture for Spatio-Temporal Reallocation in Real-Time Operations with Delayed Effects

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Abstract. Container terminals assign each external-truck job to a yard block and an appointment time before the truck arrives. A fixed assignment, however, cannot account for congestion that emerges later. This paper proposes a reinforcement learning architecture that jointly revisits the block and the arrival time of an announced job during operation. Two independent policies decide whether to propose a reallocation and whether the destination block or time slot should accept it, and a central procedure commits only mutually accepted candidates that satisfy resource constraints. Moving one job affects not only that job but also the waiting time of later trucks, the crane work sequence, and vessel supply. To capture these delayed effects, each policy learns from cost differences between discrete-event simulation branches that start from the same state and differ in one action; online decisions use only the trained networks. We evaluate the architecture in a continuous synthetic environment whose time-of-day arrival variation follows the literature and whose demand levels are design values. Compared with no reallocation, the architecture reduced total operating cost by 14.7% over 28 days, and it achieved a lower cost than a rule-based policy with the same spatio-temporal action range on 20 of 28 days ($p = 0.036$). Arrival-time adjustments accounted for 91.3% of the reduction, and the four most congested days for 90.5%. These results suggest that spatio-temporal reallocation may be more useful as a selective response to observed congestion than as a continuous intervention.

Keywords: Smart port, Container terminal, Truck appointment system, Yard block reallocation, Counterfactual cost learning, Reinforcement learning

1 Introduction

Container throughput continues to grow, but terminal land and equipment cannot expand at the same rate, and the resulting pressure appears as congestion in yard and gate operations. Studies of port efficiency show that automation, intelligence, and environmental design affect performance through different channels, so actual gains depend on how a terminal translates observed information into operational decisions [1]. Two research directions address this. On the temporal side, truck appointment systems spread arrival peaks and let terminals and carriers coordinate arrival times while retaining their own costs [2, 3]. On the spatial side, yard-operations research improves container placement, work sequencing, and crane-interference management [4–7], and reinforcement learning has been applied to real-time yard-crane dispatching [8].

Most of these approaches assume that the assigned yard block and appointment time are fixed before the truck arrives and improve operations within that assignment. During operation, however, demand may concentrate in particular time slots or blocks while crane queues change, so an assignment that was reasonable when announced may become inefficient. This paper revisits the assignment itself: for an announced job it jointly decides *where* the job should be handled and *when* the truck should arrive. Moving one job affects later truck queues, the crane work sequence, and vessel supply, so the immediate cost of reallocation cannot capture its effect; difference rewards and multi-agent counterfactual baselines address exactly this by comparing one action with an alternative [9, 10]. The destination must also decide whether it can absorb the job, separately from any benefit to the proposing side, so we use independent proposal and acceptance policies while a central procedure checks mutual consent and feasibility. Fig. 1 summarises the online decision path and the offline counterfactual learning procedure.

This study makes three contributions: (i) a post-assignment spatio-temporal reallocation architecture with independent proposal and acceptance policies and a central commitment step; (ii) simulator-based counterfactual cost learning that keeps expensive branch simulation offline and leaves only small networks and constraint checks online; and (iii) an evaluation that separates the effect of the action range from the effect of learning, and that reports both how often and by how much the learned policy is cheaper. The architecture is stated for real-time operational systems in which an assignment can be revised while its consequences appear only later, and it is instantiated and evaluated here in a container-terminal environment. Sect. 2 reviews related work, Sect. 3 presents the architecture and the learning signal, Sect. 4 describes the environment and the evaluation protocol, and Sect. 5 reports the results.

2 Related Work

Truck-appointment studies reshape demand in time: appointment windows relieve gate congestion by redistributing arrivals rather than adding capacity [2], distributed formulations let carriers and the terminal coordinate schedules while each keeps its own cost [3], and time-of-day models built from terminal records show that arrivals are far from uniform over a day [11]. Yard-operations studies reshape demand in space: crane work sequencing reduces truck waiting [4], multi-crane interference extends the scheduling problem [5], block length and crane deployment govern terminal-wide performance under long simulations [6], and the relative standing of dispatching rules depends on terminal conditions [7].

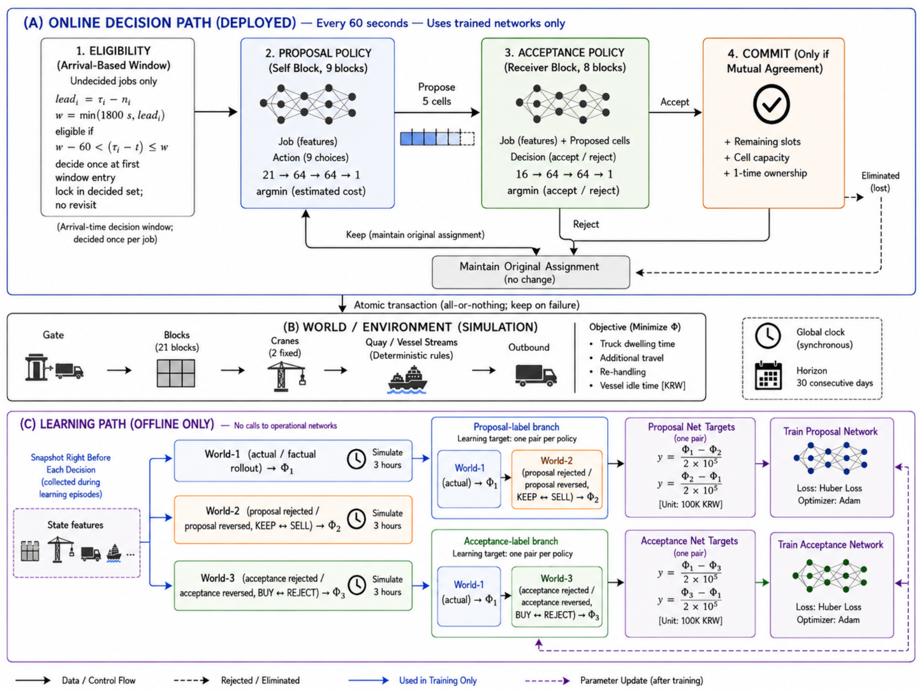


Fig. 1: The proposed architecture. (a) Every 60 s the operational path checks eligibility, proposal, acceptance, and commitment; failure of mutual consent or capacity keeps the original assignment. (b) The discrete-event environment links the gate, 21 blocks, two yard cranes per block, and vessel operations to calculate the cost Φ. (c) Offline learning compares up to three worlds from the same state and random stream over 3 h and builds separate labels for the proposal and acceptance cost networks.

Both streams optimise operations *after* the block and the appointment have been assigned; efficiency studies suggest that the decision layer that produces those assignments is itself a lever [1].

Reinforcement learning has entered this domain at the execution layer: a method based on proximal policy optimisation schedules yard cranes in real time across multiple blocks [8]. We address the preceding assignment layer, where the block and the arrival time are selected. When several actors decide jointly, the future cost is a joint outcome and the contribution of one action is hard to isolate. Difference rewards construct a payoff that is aligned with the global objective yet sensitive to the individual action [9], and counterfactual multi-agent baselines compare one actor's action with an alternative while holding the rest fixed [10]. We follow this idea, but the baseline is produced by re-simulating the terminal rather than by a learned critic.

Two modelling choices follow. First, the networks do not output action probabilities; each maps a state and one candidate action to a cost score, and the policy selects the lowest score. The score is not an absolute cost: labels are centred within each counterfactual pair (Sect. 3.4), so a network learns what one action costs *relative to* the action it was compared against. The method therefore learns a value-like quantity rather than a policy; unlike deep Q-networks, which bootstrap from their own estimates [12], the target here is a Monte-Carlo cost difference measured over a fixed horizon. Second, the approximator is a small multilayer perceptron. Feedforward networks are general approximators [13], and the input structure supports the choice. The delayed effect enters through the label, which is a forward simulation over a fixed horizon, rather than through the policy input: each candidate is scored from a fixed-length vector of the current state, candidates are scored independently, and the number of labels is small.

3 Proposed Architecture

3.1 Notation, Actions, and Policies

The decision interval is $\delta = 60\,\mathrm{s}$. The current assignment of job i is $z_i = (b_i, \tau_i)$, where b_i is the block and τ_i the announced arrival time. A reallocation action $a_i = (b'_i, \Delta_i) \in \mathcal{A}_i(t)$ gives $z_i(a_i) = (b'_i, \tau_i + \Delta_i)$; keeping the assignment is $a_i^0 = (b_i, 0)$. Block reallocation has $b'_i \neq b_i$ and arrival-time adjustment has $\Delta_i > 0$, and a candidate may change both. Outbound jobs are excluded from block reallocation because the retrieved container has a fixed location. Let n_i be the announcement time, g_i the gate-in time, and r_i an indicator of prior review. A job becomes eligible not when it is announced but when its arrival comes within a review window $w_i = \min(W,\ \tau_i - n_i)$ of the present, with $W = 1800\,\mathrm{s}$. This delays the decision until the latest feasible point before arrival, so the policy uses a more recent observation of terminal conditions:

$$\mathcal{E}_t = \{i \mid w_i - \delta < \tau_i - t \leq w_i,\ g_i > t,\ r_i = 0\}. \tag{1}$$

Each job therefore enters exactly one decision interval and is never revisited. For a job announced at least W ahead of its arrival the decision falls exactly W before that arrival, about 46 minutes later than the median announcement; a job announced with a shorter lead is reviewed immediately. A job announced no earlier than its own arrival has $w_i = 0$ and is never eligible; such jobs are 14.1% of the demand, and the architecture leaves them untouched. The proposal policy scores each candidate from the block state and job-action features o_i^P and selects the lowest score, while the acceptance policy compares acceptance with rejection from the proposal message and the destination state o_i^A:

$$\begin{aligned} \hat{a}_i &= \arg\min_{a \in \mathcal{A}_i(t)} \hat{c}_\theta^P(o_i^P, a), \qquad p_i = \mathbf{1}[\hat{a}_i \neq a_i^0], \\ q_i &= \mathbf{1}\left[\hat{c}_\psi^A(o_i^A, \hat{a}_i, \mathrm{acc}) \leq \hat{c}_\psi^A(o_i^A, \hat{a}_i, \mathrm{rej})\right]. \end{aligned} \tag{2}$$

If the target resource has no remaining capacity, the request is rejected before the network is called. The mutually accepted candidates form $\mathcal{J}_t = \{i \in \mathcal{E}_t \mid p_i q_i = 1\}$.

3.2 Central Commitment

The central procedure creates no new candidates. It considers only $\mathcal{J}_t$, sorted by the acceptance policy's predicted cost with lexicographic tie-breaking. With $x_i \in \{0,1\}$ indicating commitment, $F_b(t)$ the free slots in block b, and $K_k(t)$ the remaining quota of time slot k, it enforces

$$\begin{aligned} x_i \le p_i q_i, \qquad & \sum_{i\in\mathcal{J}_t} x_i \mathbf{1}[b'_i = b] \le F_b(t), \\ \sum_{i\in\mathcal{J}_t} x_i \mathbf{1}[k(i) = k] \le K_k(t), \qquad & \sum_{t\in\mathcal{T}} \mathbf{1}[i \in \mathcal{E}_t] \le 1. \end{aligned} \tag{3}$$

Block capacity is enforced in every experiment; when time-slot quota data are unavailable we set $K_k(t) = \infty$, so the arrival-time results represent the potential range of adjustment rather than an executable schedule. All committed assignments are applied together at the end of the decision epoch, so an earlier commitment cannot change the state seen by another candidate in the same epoch.

3.3 Cost Networks

Both $\hat{c}^P_\theta$ and $\hat{c}^A_\psi$ are multilayer perceptrons with two 64-unit hidden layers and ReLU, with weights shared across blocks: $21 \to 64 \to 64 \to 1$ for proposal and $16 \to 64 \to 64 \to 1$ for acceptance, or fewer than 11,000 parameters in total. Inputs include current waiting, inventory, remaining crane work, the public expected arrival time, route difference, and announced volume near the target time; unrealised arrival and completion times and simultaneous responses from other accepting parties are excluded. The output is one pair-centred cost score per candidate, so the length of the candidate list may change without retraining. Mean subtraction inside a pair removes a constant that is common to both actions and therefore leaves the ranking in (2) unchanged.

3.4 Counterfactual Cost Labels and Training

For each sampled decision we clone the state and the random stream just before branching and run up to three worlds for $H = 10{,}800\,\mathrm{s}$: $\phi^F_i = \Phi_H(\text{observed})$, $\phi^P_i = \Phi_H(\text{proposal reversed})$, and $\phi^A_i = \Phi_H(\text{acceptance reversed})$. The counterfactual branch reverses the policy's observed decision: if the proposal policy proposed a change the alternative keeps the assignment, if it kept the assignment the alternative commits the cheapest candidate under the current weights, and the acceptance alternative swaps acceptance and rejection. For each policy the two actions of the pair are centred on their mean and divided by KRW 100,000,

$$\bar{\phi}^P_i = \frac{\phi^F_i + \phi^P_i}{2}, \quad y^P_{i,F} = \frac{\phi^F_i - \bar{\phi}^P_i}{10^5}, \quad y^P_{i,P} = \frac{\phi^P_i - \bar{\phi}^P_i}{10^5}, \tag{4}$$

that is, $\pm$ half the pairwise cost difference; the same centring is applied to (ϕ_i^F, ϕ_i^A). The reference point is therefore the pair itself rather than a fixed action.

We sample at most 64 decisions per day, split them 80/20 into training and validation, and optimise with Adam (3×10^{-4}) and Huber loss ($\beta = 1.0$), which prevents a few very large counterfactual differences from dominating the fit [17]. Exploration decreases linearly from 0.50 to 0.05 over the 30 training iterations. The labels are not a fixed dataset: every pair branches from the action the current policy actually took, so both the labelled states and the alternative they are compared against move as the weights change, and each iteration's labels are discarded after its update. In this respect the scheme is on-policy Monte-Carlo advantage estimation with a simulator-generated baseline; it uses neither bootstrapping nor a policy gradient. Training both networks for 30 iterations took about 18 hours of wall-clock time. During evaluation the weights are frozen, exploration is disabled, and no counterfactual branch is called, so online operation costs one forward pass per candidate.

4 Experimental Environment and Evaluation Protocol

We evaluate the architecture in a literature-based synthetic environment rather than in the operating record of a particular terminal. The studies of Sect. 2 provide the modelling ranges—20 blocks [7], multiple cranes per block [5], continuous multi-week operation [6], strong time-of-day variation in truck arrivals [11], three vessel classes [7], and a reported range of crane handling times [6]—and Table 1 lists the values used within them. Following the practice of separating measured ranges from synthetic parameters and identifying the synthetic component explicitly [14], we state which values are design choices: the daily demand levels, the shape of the arrival curve, the counterfactual horizon, and two of the four cost coefficients.

The total cost over $[d, d+1)$ is the sum of four terms, where h_i is the dwell time of truck i, m the additional crane working time, R the number of rehandles, and T_v the policy-related idle time of vessel v:

$$\begin{aligned} \Phi_d &= C_{wait,d} + C_{move,d} + C_{rehandle,d} + C_{vessel,d}, \\ C_{wait} &= \sum_i 40{,}000\{h_i + \max(0, h_i - 1)\}, \qquad C_{move} = 60{,}000\, m, \\ C_{rehandle} &= 2{,}000\, R, \qquad C_{vessel} = \sum_v 2.99\, GT_v T_v. \end{aligned} \tag{5}$$

The base monetary scale is derived from the safe trucking freight rate [15], whereas the conversion to an hourly dwell-cost coefficient is a design choice. Structural vessel time that the policy cannot change is recorded separately. Waiting dominates the total: C_{wait} accounts for 96.4%, against 2.2% for rehandling, 1.2% for vessel idling, and 0.1% for crane movement, so conclusions based on

Table 1: Experimental setup. Values marked *design* are choices of this study; the others lie within ranges reported in the cited work. Job announcement times follow $f(t) = 0.38\,U(0,24) + 0.62\sum_{m=1}^{3} w_m \mathcal{N}(t;\mu_m,\sigma_m^2)$ with (μ_m,σ_m,w_m) equal to $(10,1.5,.317)$, $(15,2.5,.633)$, and $(21,1,.05)$, time in hours.

Group	Setting	Value
Yard	Blocks, cranes per block	21, 2
	Slots per block, initial occupancy	1,440, 65%
Time	Decision interval δ	60 s
	Review window W	1,800 s
	Counterfactual horizon H (design)	10,800 s
Demand	Announcement lead	median 1.27 h (14.1% ≤ 0)
	Daily levels (design)	3,500, 5,000, 7,500, 12,500, 15,000
	Draw probabilities (design)	0.30, 0.30, 0.25, 0.10, 0.05
	Arrival curve (design)	uniform base 38% + 3 Gaussians
Vessels	Classes (GT), calls/day, crane rate	50/100/150k, 3.3, 27.5 moves/h
Cost	Truck dwell	KRW 40,000/h (twice beyond 1 h)
	Crane move, rehandle (design)	KRW 60,000/h, KRW 2,000/event
	Vessel idling	KRW 2.99 per GT-hour [16]
Learning	Networks (proposal, acceptance)	21/16 $\to$ 64 $\to$ 64 $\to$ 1
	Iterations, decisions per day	30, ≤ 64
	ε, optimiser	0.50 $\to$ 0.05, Adam 3×10^{-4}
Evaluation	Days, compared, seed	30, middle 28, 9,900,950
	Policy at evaluation	frozen weights, $\varepsilon = 0$

Φ are driven mainly by the truck dwell coefficient. Congestion changes the cost structure as well as its level: waiting is 83.2% of cost at 3,500 trucks per day and 99.1% at 15,000, and the daily 90th-percentile turn time ranges from 0.79 h on the lightest days to 7.49 h on the heaviest.

The evaluation uses a 30-day continuous run with random seed 9,900,950, which was not used in training; the middle 28 days are compared and the first and last days only connect the continuous boundaries. All policies share the same truck, vessel, and service random streams on a given day, following the common-random-number method [18]. Implementation checks confirm that proposal and acceptance use separate models, that one-sided consent never produces a commitment, that realised future times are excluded from policy inputs, and that state continues across day boundaries; frozen-policy evaluation made zero counterfactual calls, and a branch that changed no action reproduced the main run at all four sampled points. We compare the learned policy with three groups: its own ablations, which allow only arrival-time adjustment or only block reallocation; the model after the first of the 30 training iterations, which isolates the effect of the remaining iterations; and six rule-based policies, four of which reallocate blocks (least slack, first-come-first-served, shortest processing time, net gain) while two transfer the *least-loaded* criterion to the temporal axis and therefore match the spatio-temporal action range of the learned policy.

5 Results

5.1 Cost Reduction and Paired Daily Comparison

Over the 28 compared days, total cost was KRW 10.00 bn without reallocation and KRW 8.52 bn with the learned policy, a reduction of KRW 1.471 bn or 14.7% (Table 2). Days are treated as independent comparison units, and we report two paired tests because they answer different questions. A two-sided sign test counts only the days on which one policy is cheaper, so it is blind to how large each day's difference is; a Wilcoxon signed-rank test ranks the differences by magnitude before summing them, so a policy that wins on fewer days but by a wider margin is scored differently by the two tests. The distinction matters here because the learned policy acts selectively. With 28 days and no tied differences, both tests are computed exactly.

The learned policy costs less than the rule with the same block-and-time action range on 20 of 28 days (sign test $p = 0.036$, signed-rank $p = 0.015$), and less than no reallocation on 26 of 28 days ($p < 0.001$). It costs less than the first-iteration model on 18 days, but this difference is not statistically significant under either test ($p = 0.185$ and $p = 0.164$). At the 5% level the two tests agree on every row of Table 2, although the signed-rank p is the smaller of the two in

Table 2: Main comparison. "Reduction" is the 28-day total against no reallocation; positive means lower cost. The paired columns compare the learned policy with that row's policy on the same days, and p values are two-sided and exact. Values below 0.05 are in bold. Against no reallocation the two days on which the learned policy is more expensive cost KRW −17.1 m and KRW −0.5 m, whereas each of the four congested days exceeds KRW 290 m.

Policy	Actions	Reduction	Learned cheaper	Sign p	Rank p
Learned policy	block+time	**+14.72%**	—	—	—
Learned, time only	time	+13.44%	—	—	—
Learned, block only	block	−1.31%	—	—	—
First-iteration model	block+time	+7.97%	18/28	0.185	0.164
Rule: least-loaded slot	time	+3.41%	—	—	—
Rule: least-loaded block+slot	block+time	+1.51%	20/28	**0.036**	**0.015**
Rule: least slack	block	+0.53%	20/28	**0.036**	**<0.001**
Rule: first-come-first-serve	block	+0.17%	22/28	**0.004**	**<0.001**
Rule: shortest processing time	block	+0.05%	21/28	**0.013**	**0.006**
Rule: net gain	block	−0.48%	21/28	**0.013**	**0.003**
No reallocation	—	0.00%	26/28	**<0.001**	**<0.001**

most rows, which is consistent with an effect concentrated in a small number of high-cost days.

The temporal comparison shows why both tests are needed. Restricted to arrival-time adjustment, the rule that transfers the least-loaded criterion to the temporal axis reduces cost by 3.41% relative to no reallocation, so the temporal action provides a measurable benefit even without learning. Against that rule the learned time-only policy is cheaper on only 8 of 28 days, which the sign test reports as a difference in favour of the rule ($p = 0.036$); yet the signed-rank test finds no difference ($p = 0.399$) and the learned policy's 28-day total is KRW 1.00 bn lower. This is the one comparison in which the two tests disagree, and reporting only how often a policy wins would invert the conclusion.

5.2 Demand Level and the Decomposition of Actions

Fig. 2 decomposes the reduction by demand level and by action range. The four days with 12,500 or 15,000 trucks account for 90.5% of the total, which supports developing the architecture towards a policy that *intervenes selectively when congestion is observed* rather than continuously. Restricting the policy to one action type at a time separates the two contributions: arrival-time adjustment alone gives KRW 1.344 bn and block reallocation alone −KRW 0.131 bn, so their interaction accounts for the remaining KRW 0.258 bn of the KRW 1.471 bn total. Arrival-time adjustment therefore explains 91.3% of the reduction. This share should be read together with the cost structure: truck waiting is 96.4% of total cost here, so arrival-time adjustment acts directly on the dominant component, and the relative standing of spatial and temporal actions may differ under a different cost mix. Block-only reallocation saves KRW 0.124 bn under smooth and normal demand but raises cost at the highest demand, and the policy should therefore adjust its use of spatial and temporal actions by demand level.

5.3 Effect of Learning and Sensitivity to the Dispatching Rule

Across the 30 training iterations the proposal network's Huber loss decreases from 0.079 to 0.029 on training decisions and from 0.268 to 0.119 on validation decisions; the acceptance network decreases from 0.171 to 0.027 and from 0.315 to 0.136 (means over the first and last five iterations, targets in units of KRW 100,000). The decline in validation loss indicates that the learned mapping generalises to held-out decisions, although the final validation loss remains about four times the training loss; with about 56 labelled decisions per iteration, this supports keeping the model small. The share of decisions whose counterfactual difference is exactly zero falls from 51.4% to 37.6% as exploration decreases, and training used 4,671 counterfactual worlds in total (Fig. 3).

The first-iteration model cost KRW 9.20 bn over the same days, so the remaining 29 iterations added KRW 0.675 bn, or 45.9% of the total reduction. The trained model is cheaper on only 18 of 28 days ($p = 0.185$), and the four congested days account for 94.6% of this increment, and the trained model is cheaper on all four. A stratified bootstrap over the daily differences with 20,000

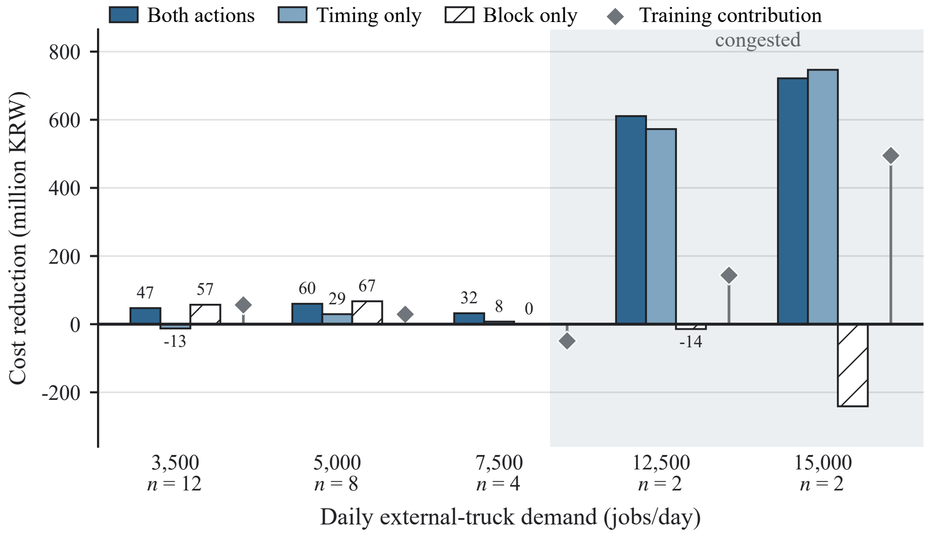
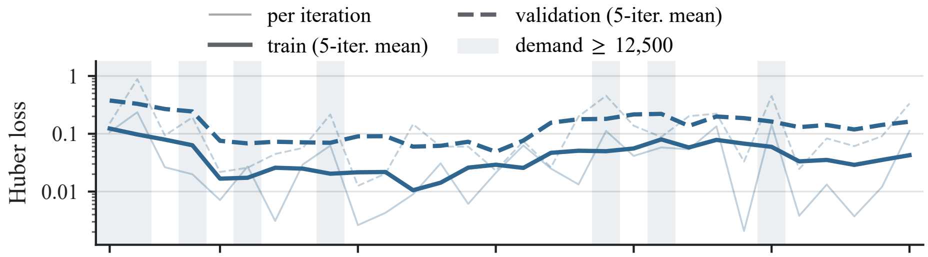


Fig. 2: Cost reduction by demand level, decomposed by action range, in million KRW. The shaded band marks the two congested levels; because those are more than ten times larger than the rest, the value is printed next to each short bar, and n is the number of days observed. Bars measure a restricted policy against no reallocation, whereas the diamond is the learned policy minus the first-iteration model and therefore carries a different mark.

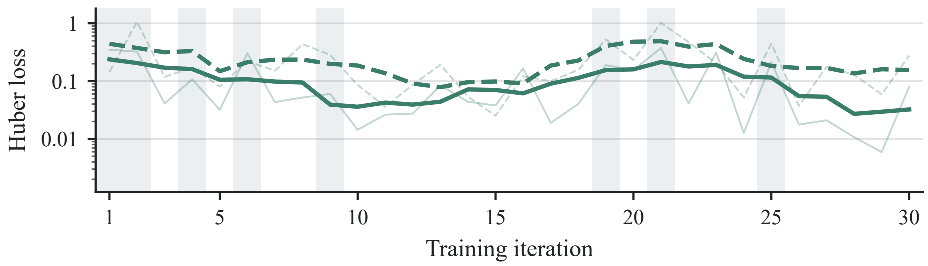
resamples gives a 95% interval of $[-32, -17]$ m per day for the mean difference, which excludes zero: the learning effect is large in total and concentrated, but not frequent enough for the day-level sign test at a conventional significance level. Establishing it more firmly requires more congested days rather than more days overall. Finally, the ranking of crane-dispatching rules changes with demand —the shortest processing time rule is 22–25% more expensive under smooth demand but the cheapest from 7,500 trucks upward—so the reallocation effect should also be tested under other rules.

6 Conclusion

This paper proposed a reinforcement learning architecture that revisits the yard block and the arrival time of an already announced external-truck job during operation. Independent proposal and acceptance policies and a central commitment step connect distributed appointment coordination [3] and yard-crane operation [4, 5] into one spatio-temporal assignment problem, and each policy learns from counterfactual cost differences measured over a fixed horizon rather than from an immediate movement cost; this brings multi-agent counterfactual learning [9, 10] into the operating decision. Because the expensive branch simulation is used only during offline training, online operation requires only small



(a) Proposal network



(b) Acceptance network

Fig. 3: Training and validation loss over the 30 iterations, with the per-iteration value drawn faintly beneath a five-iteration mean; the vertical axis is logarithmic. Shading marks the iterations in which the demand level drawn was 12,500 or above, so much of the visible fluctuation follows that draw rather than the fit. Exploration decreased linearly from 0.50 to 0.05.

cost networks and constraint checks. In the synthetic environment the architecture reduced cost by 14.7% against no reallocation and was cheaper than a rule with the same action range on 20 of 28 days; the reduction came mainly from arrival-time adjustment on congested days. Real-time congestion information was therefore turned into assignment actions and not only into monitoring. Because a rule can be cheaper on more days while the learned policy is cheaper in total, both the frequency of lower-cost outcomes and the aggregate reduction are needed for interpretation.

These results establish the architecture in a literature-based synthetic environment rather than estimating performance at a particular terminal, and the concentration of the effect on congested days suggests operating the policy selectively rather than continuously. Future work will extend the framework along three axes: terminal-specific arrival, demand, and cost data; executable appointment quotas and alternative acceptance structures, including the value of the acceptance veto; and robustness across counterfactual horizons, cost structures, and crane-dispatching rules. These extensions retain the observation–proposal–acceptance– commitment–execution structure proposed here while defining the conditions under which spatial and temporal reallocation are most effective.

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